Sinchana Vinayak Bhat

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Education

RV INSTITUTE OF TECHNOLOGY & MANAGEMENT **Sep. 2023 – July 2027**
Bachelor of Engineering in Information Science Engineering *CGPA: 9.25*

Technical Skills

Languages: Python, Java, SQL, C, JavaScript
Artificial Intelligence: Machine Learning, Deep Learning, Computer Vision, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)
Frameworks & Libraries: PyTorch, TensorFlow, MONAI, Scikit-learn, LangChain, SentenceTransformers
Backend & Databases: Node.js, Express.js, FastAPI, MongoDB, MySQL, REST APIs
Tools: Git, Linux, Jupyter Notebook, Google Colab
Core CS: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Projects

Self-Supervised 3D Swin-UNETR for Brain Tumor Segmentation and Explainable Classification
Python — PyTorch — MONAI — Swin-UNETR — Grad-CAM

- Implemented and fine-tuned a self-supervised 3D Swin-UNETR pipeline for brain tumor segmentation using the BraTS MRI dataset.
- Developed a segmentation-guided classification framework with Grad-CAM based explainability for improved clinical interpretability.
- Processed multi-modal MRI volumes (T1, T1ce, T2, and FLAIR) using MONAI and PyTorch-based preprocessing pipelines.
- Achieved Dice scores of approximately **0.80 (WT), 0.57 (TC), and 0.42 (ET)** on the BraTS validation dataset.

Intelligent Offline Research Assistant using Retrieval-Augmented Generation *Python — Streamlit — GPT4All — ChromaDB — SentenceTransformers*

- Engineered an offline Retrieval-Augmented Generation (RAG) system for answering entrepreneurship-related queries using custom PDF knowledge bases.
- Built semantic search pipelines using ChromaDB and SentenceTransformers for accurate contextual retrieval.
- Integrated GPT4All local LLM with multi-document support, persistent vector storage, and interactive Streamlit interface.
- Enabled document-based conversational AI with fully offline inference for enhanced privacy and accessibility.

Secure Military Communication System *Java — Java Swing — JCA — MySQL*

- Designed a secure communication platform combining Huffman compression, AES encryption, and RSA key exchange.
- Implemented real-time visualization of Huffman tree generation, encoding, and decoding processes.
- Improved transmission efficiency while ensuring confidentiality and secure key management.

Quantum-ML Hybrid Energy Demand Forecasting *Python — TensorFlow — Streamlit — Plotly*

- Developed an AI-driven forecasting model using historical energy consumption and weather datasets.
- Implemented QLSTM networks to capture temporal dependencies in electricity demand patterns.
- Built interactive dashboards using Streamlit and Plotly for visualization and prediction analysis.

Research Experience

- **Machine Learning Model for Predicting the Ranking Number of Engineering Courses**
  - Developed machine learning models to predict engineering course rankings using historical educational datasets.
  - Status: Under Review.
- **Machine Learning-Based TES Collapse Moment Estimation for Angled Pipe Bends under Internal Pressure and Out-of-Plane Bending**
  - Applied machine learning techniques for structural engineering prediction problems.
  - Status: Under Review.

Hackathons & Leadership

- Team Lead, **Fantomcode Hackathon 2023**
  - Led the development of EmpHeal, an AI-assisted Mental Health Advisor.
- Volunteer Tutor, **U&I Foundation**
  - Mentored school students through academic support initiatives.
- Member, **AKSA Drama Club**

Certifications

- IBM SkillsBuild — Introduction to Artificial Intelligence
- Infosys Springboard — Advanced Java
- Cisco Networking Academy — Networking Basics

Coding Profiles

LeetCode: leetcode.com/u/Sinchana__BHAT/
Strivers A-Z DSA Sheet: takeuforward.org/profile/sinchana_bhat17